

MIR MEDICAL INTELLIGENCE

Laboratory-First Clinical Intelligence Research Program

Research Proposal, Methodology, Data Acquisition Plan, ML/DL Architecture, Federated Learning Track, Agentic AI Track, and Multi-Year Roadmap

Prepared: 5 September 2026

Primary implementation language: Python | Secondary analytical language: R

Scope: non-imaging laboratory and clinical data; CT/X-ray intentionally excluded from the initial research program

Core proposition: Investigate whether routine laboratory measurements and longitudinal laboratory trajectories contain sufficient signal to detect physiological anomalies and estimate future disease risk before a formal diagnosis becomes obvious, while keeping the system as clinician decision support rather than an autonomous diagnostic authority.

Important safety and research position

This document proposes a research program, not a medical device specification for immediate clinical use. Any model that reaches clinical use would require appropriate clinical evidence, quality management, risk management, regulatory assessment, cybersecurity, human factors evaluation, and clinician oversight. In the European Union, software intended to provide information used for diagnostic or therapeutic decisions may fall under MDR Rule 11, and classification depends on intended use and the consequences of decisions supported by the software. [16][17]

1. Executive Summary

MIR Medical Intelligence (MIR-MI) is proposed as a research platform for laboratory-first clinical intelligence. The program will study whether routine, non-imaging data can support three related tasks: (1) detection of clinically meaningful physiological anomalies; (2) characterization of longitudinal trajectories and early deterioration; and (3) prediction of future disease outcomes, including selected cardiovascular, metabolic, renal, hepatic, hematologic, inflammatory and oncologic outcomes.

The research intentionally avoids using CT scans and X-rays as model inputs in the first program phase. This constraint creates a clear scientific question: how much clinically useful information is contained in ordinary laboratory data alone, and how much additional signal is obtained by modeling change over time rather than isolated values?

The program should be built as a sequence of increasingly difficult research tracks. Track A establishes data quality and physiological anomaly detection. Track B develops disease-specific prediction models. Track C introduces temporal and survival models. Track D studies external validity and dataset shift. Track E develops federated learning so institutions can collaborate without centralizing patient-level data. Track F introduces tightly constrained agentic AI as an orchestration and evidence-retrieval layer, never as an unconstrained autonomous diagnostician. Track G addresses prospective silent validation and clinical workflow evaluation.

The initial data strategy is deliberately heterogeneous. NHANES is best suited to population-level laboratory distributions and reference variation; MIMIC-IV and eICU provide longitudinal hospital/critical-care EHR data and outcomes; All of Us offers broad EHR and measurement data through its secure Researcher Workbench; UK Biobank is an important long-term option but currently reports that new applications are paused while platform changes are made, with new applications expected in late 2026; N3C is a further advanced option with controlled access requirements. [1][2][3][4][5][6]

The project will use reproducible cohort definitions, patient-level and temporal splits, external validation, calibration, subgroup analysis, uncertainty reporting, leakage controls, experiment tracking, model cards, and research logs. Reporting will be structured around TRIPOD+AI for prediction models, PROBAST+AI for risk of bias/applicability, and STARD/STARD-AI principles when a study is framed as diagnostic accuracy rather than future-risk prediction. [11][12][13]

Program objective

Develop and rigorously evaluate a laboratory-first clinical intelligence framework that can surface abnormal physiological patterns and calibrated future-risk signals from non-imaging clinical data, with every output traceable to data provenance, model version, evidence, uncertainty, and human review.

Primary deliverables

- A reproducible Python research repository with data ingestion, harmonization, feature engineering, modeling, evaluation, and report generation.
- A laboratory anomaly engine covering statistical, classical ML, and deep anomaly-detection approaches.
- A longitudinal trajectory engine that models level, slope, variability, persistence, and change points.
- A disease-risk benchmark suite spanning logistic regression, tree ensembles, neural networks, survival models, and calibrated ensembles.
- A federated-learning research environment using multiple simulated/partner nodes, followed by real multi-site studies when governance permits.
- A constrained clinical agent layer for evidence retrieval, tool orchestration, report drafting, provenance checking, and workflow support.

- A clinician-facing decision-support interface with explainability, uncertainty, data-coverage indicators, and auditability.
- A complete research record suitable for eventual manuscripts, theses, conference submissions, model cards, and regulatory evidence packages.

2. Research Framing and Scientific Premise

2.1 Clinical problem

Routine laboratory testing is abundant in modern healthcare, but laboratory values are commonly interpreted one result at a time and inside disease-specific workflows. The proposed research asks whether multivariable and longitudinal patterns can reveal early physiological deviation that is not obvious from a single result. The goal is not to redefine what a CBC or chemistry panel clinically means; it is to determine whether machine learning can extract additional, reproducible, and clinically useful signal from combinations and trajectories of routine measurements.

2.2 Central hypothesis

H1: longitudinal combinations of routine non-imaging laboratory measurements can identify physiologically abnormal states and estimate risk of selected future clinical outcomes better than single-test or simple threshold-based approaches, provided the research controls for confounding, missingness, dataset shift, label leakage, and calibration.

2.3 Counter-hypothesis

H0: after accounting for age, sex, care setting, missingness, and known clinical variables, laboratory-only models provide little additional predictive information for certain outcomes, or their performance is too poorly calibrated and non-generalizable to support meaningful decision support. This negative result would still be scientifically valuable and should be reported.

2.4 Clinical boundaries

- The system should not claim that a laboratory pattern proves a diagnosis.
- Cancer modules should be framed as risk/signal models, not universal cancer detectors.
- Recommendations should be phrased as clinician considerations, not autonomous treatment orders.
- The system must disclose when evidence is insufficient, data coverage is poor, or the model is outside its validated population.
- Imaging and pathology may later be used as reference standards or outcomes, but they are excluded as model inputs in the laboratory-first track.

3. Clinical Questions and Research Questions

3.1 Clinical questions

1. Can routine laboratory patterns identify physiological abnormalities beyond simple reference-range flagging?
2. Can persistent or progressive changes in laboratory values reveal early deterioration before a formal diagnosis is recorded?
3. Which laboratory combinations are associated with inflammatory, metabolic, renal, hepatic, hematologic, cardiovascular, infectious, endocrine, or nutritional states?

4. Can longitudinal lab trajectories distinguish transient abnormalities from persistent patterns?
5. Can routine laboratory data identify elevated future risk for selected diseases without using CT/X-ray images as predictors?
6. For selected cancers, can non-imaging laboratory data provide a measurable early risk signal within defined time windows?
7. How much does performance improve when historical trajectories are added to current values?
8. Can calibrated ML outputs be translated into clinician-reviewable evidence rather than opaque risk scores?

3.2 Primary methodological questions

1. Do multivariable laboratory models outperform reference-range rules and simple statistical baselines?
2. Do temporal features improve discrimination and calibration compared with cross-sectional features?
3. Which model family is most robust under missingness, class imbalance, and distribution shift?
4. Does an ensemble materially improve performance over the best single model after proper calibration?
5. Which anomaly-detection method is most stable across datasets?
6. How much performance degrades under external validation at another institution or care setting?
7. Can federated learning preserve or improve external validity while keeping raw patient-level data local?
8. Can agentic orchestration improve clinician workflow without introducing unsafe autonomous decision-making?
9. Which explanation methods are stable enough to be useful to clinicians?
10. What evidence threshold is required before a prediction is exposed in a clinical interface?

3.3 Secondary questions

Secondary analyses should examine subgroup performance, calibration drift, data missingness as signal, laboratory reference-range heterogeneity, site-specific measurement practices, model compression, uncertainty estimation, and the trade-off between sensitivity and false alerts.

4. Research Tracks

Track	Purpose	Primary data	Methods
A. Data & Normality	Build canonical laboratory data and estimate physiological variation.	NHANES + synthetic data	Schema, harmonization, missingness, reference ranges
B. Unsupervised Anomaly	Detect unusual multivariable and longitudinal profiles without disease labels.	NHANES + MIMIC-IV	Isolation Forest, LOF, One-Class SVM, AE/VAE
C. Disease Prediction	Predict selected well-defined outcomes.	MIMIC-IV / eICU / All of Us	LR, RF, XGBoost, LightGBM, CatBoost, MLP
D. Temporal Risk	Model trajectories and time-to-event outcomes.	MIMIC-IV / All of Us / UKB later	GRU, LSTM, Transformer, Cox, RSF, DeepSurv
E. Oncology	Test selected cancer-risk hypotheses without imaging inputs.	All of Us / UKB / N3C where eligible	Survival, gradient boosting, temporal DL
F. External Validation	Measure transportability across datasets and sites.	eICU, All of Us, partner data	Temporal + geographic/site validation
G. Federated Learning	Train without moving raw patient data.	Synthetic nodes first; partner sites later	FedAvg, FedProx, SCAFFOLD, personalization
H. Agentic AI	Orchestrate evidence/reasoning tools under strict controls.	All validated models	Tool-use agents, RAG, verifier agents
I. Silent Prospective	Run predictions without	Clinical partner	Prospective observational /

Track	Purpose	Primary data	Methods
Evaluation	influencing care.		silent study
J. Translation	Evaluate human-AI workflow and clinical utility.	Clinical partner	Human factors, decision curves, usability, safety

5. Dataset Strategy and Acquisition Roadmap

5.1 Dataset portfolio

Dataset	What it provides	Access route	Primary role	Ref.
NHANES	Public CDC survey/laboratory data; excellent for population variation and lab distributions.	Free/public download; follow CDC data-use and survey-weight guidance.	Baseline laboratory profiling, anomaly detection, population reference analyses.	[3][7]
MIMIC-IV	Credentialed de-identified EHR/hospital data with labs and clinical outcomes.	PhysioNet credentialing + required CITI training + DUA.	Primary clinical-development dataset; temporal trajectories and outcomes.	[1]
eICU-CRD	Multicenter critical-care database.	Credentialed access + human research training + DUA.	External validation and transportability.	[2]
All of Us	Large research program with EHR, measurements and other participant data; OMOP-standardized EHR.	Researcher Workbench registration; institutional DURA; tier-specific access.	Large-scale validation, broader population, oncology/cardiometabolic research.	[4][5][6]
UK Biobank	Large longitudinal resource with extensive phenotype, biomarkers and outcomes.	Application + scientific review + institutional process + fee/MTA after approval. New applications currently paused; late-2026 reopening expected.	Advanced validation and long-term risk studies.	[8][9]
N3C	Large U.S. EHR research enclave with multiple access levels.	N3C registration, enclave, DUA, training, data-use request; non-U.S. researchers limited to specified tiers.	Advanced external validation and federated-methodology research where eligible.	[10]
Synthea / SyntheticMass	Realistic synthetic EHR generator and synthetic population.	Open/free synthetic data and local generation.	Software development, schema tests, FL prototypes, agent evaluation.	[14]
SEER	Population cancer registry incidence, tumor characteristics and survival; not a direct laboratory dataset.	Data-access route depends on product.	Cancer outcome definitions/context; use only where linkage or compatible cohort permits.	[15]

5.2 Recommended acquisition order

1. Start with Synthea for application schema, import/export, FHIR handling, feature engineering tests, and federated-learning simulation. It is synthetic and should never be treated as a substitute for clinical evidence. [14]
2. Acquire NHANES public files and build the laboratory canonicalization layer, beginning with CBC and broad laboratory panels. The current 2021-2023 laboratory repository includes CBC and numerous additional laboratory files. [3][7]

3. Complete PhysioNet credentialing, CITI Data/Specimens Only Research training, and the MIMIC-IV data-use agreement. Do not copy or redistribute restricted patient-level files in your public repository. [1]
4. Add eICU as a deliberately external dataset rather than mixing it into the training set. Use it to test transportability. [2]
5. Apply to All of Us when institutional access is available. The Researcher Workbench is a secure cloud environment with JupyterLab and RStudio options and OMOP-standardized EHR data. [4][5][6]
6. Prepare a UK Biobank application package for the late-2026 reopening window if the intended research requires its biomarkers and outcomes. [8][9]
7. Consider N3C only when the research question and institutional eligibility justify its governance overhead. [10]

5.3 Acquisition checklist

- Create a data-source register: owner, dataset/version, access date, license/DUA, intended use, restrictions, citation, and retention period.
- Record exactly which release/version was used for every experiment.
- Keep raw restricted data outside Git and public artifact stores.
- Store only derived non-identifying aggregates or synthetic examples in the public repository.
- Document institutional approvals, IRB/ethics determinations where required, data-processing roles, and security controls before adding real institutional data.
- Create a data dictionary before model training, including source field, unit, vocabulary, reference range, missingness meaning, and transformation history.

6. Data Standardization and Clinical Data Model

6.1 Canonical internal schema

The internal model should preserve original observations and create separate normalized features. LOINC is appropriate for identifying laboratory and clinical observations; its current guidance explicitly emphasizes universal identifiers for pooling results across systems. OMOP CDM should be used as the conceptual target for broader interoperability and multi-site observational research. [20][21]

Field group	Examples
Identity	patient_id, encounter_id, observation_id
Clinical concept	LOINC code, local code, OMOP concept_id
Measurement	value, unit, UCUM unit, specimen
Reference context	reference_low, reference_high, reference_text
Time	collection_time, result_time, encounter_start, encounter_end
Source	dataset, site, laboratory source, instrument if available
Quality	missing_reason, plausibility_flag, duplicate_flag, transformation_version
Outcome linkage	outcome_definition_id, index_date, event_date, censor_date

6.2 Essential laboratory domains

- Hematology: CBC and differential; reticulocytes where available.
- Inflammation/immune: CRP, ESR, ferritin and other available markers.
- Renal/electrolytes: creatinine, eGFR-related variables, BUN/urea, sodium, potassium, bicarbonate, calcium, phosphate.
- Hepatic: ALT, AST, ALP, GGT, bilirubin, albumin, total protein.

- Metabolic/cardiovascular: glucose, HbA1c, total cholesterol, LDL, HDL, triglycerides.
- Coagulation and thrombotic markers: PT/INR, aPTT, D-dimer where present.
- Nutritional/endocrine: iron, transferrin, B12, folate, vitamin D, TSH and other available markers.
- Contextual data: age, sex, BMI, blood pressure, smoking and clinical history only when the dataset and study protocol allow.

6.3 Reference ranges and units

Never reduce all laboratory results to a single hard-coded global range. Preserve the reported reference interval when available. Normalize units using a controlled unit system such as UCUM and maintain the original value/unit alongside the normalized value. Reference ranges can vary by laboratory, method, age, sex, and clinical context.

7. Research Methodologies

7.1 Study design progression

Method	Use	Key safeguard	Primary reporting
1. Exploratory/Descriptive	Characterize distributions, missingness, unit systems, reference ranges, lab ordering patterns.	No causal claims.	Data-quality report.
2. Retrospective Cohort	Define index date, predictors available before index, and outcomes after index.	Primary design for prediction.	TRIPOD+AI + PROBAST+AI.
3. Diagnostic Accuracy	Compare an AI index test against a reference standard within a defined window.	Use only when the question is diagnostic classification.	STARD/STARD-AI principles.
4. Temporal Validation	Train on earlier time, test on later time.	Tests temporal drift.	Pre-specified cutoff.
5. External Validation	Train on one dataset/site, test on another.	Tests transportability.	No test-set tuning.
6. Prospective Silent Study	Predictions generated prospectively but hidden from clinicians.	Measures real-world distribution and drift without influencing care.	Prospective observational protocol.
7. Human-AI Study	Compare clinician workflow with and without model support.	Measures usefulness, calibration of trust, false-alert burden.	Human factors / clinical study.
8. Federated Evaluation	Run the same protocol across local nodes and aggregate permissible statistics/model updates.	Tests privacy-preserving collaboration.	Federated protocol + site harmonization.

7.2 Cohort construction

1. Define the target population and inclusion/exclusion criteria before model development.
2. Define a reproducible index date or prediction window.
3. Specify the data availability window before index date, e.g., current values only, 30/90/365-day history, or all prior measurements.
4. Define outcome events and follow-up/censoring rules before examining model performance.
5. Create patient-level splits so the same patient cannot appear across training and test sets.
6. Use temporal separation where feasible; preserve a fully untouched final test set.
7. Freeze the model and preprocessing pipeline before external validation.

7.3 Leakage prevention

Label leakage is a central risk. The study must ensure that predictors are restricted to information available at or before the prediction time. This includes avoiding post-index tests, diagnoses, procedures, or measurements that were triggered by an already-suspected disease unless the study explicitly investigates that clinical workflow. Leakage checks should be automated and reviewed by a clinician/data steward before each major experiment.

7.4 Missingness methodology

Missingness should be analyzed as data-generating behavior, not only as a preprocessing inconvenience. Compare transparent imputation approaches with models that natively handle missing values, and separately evaluate whether missingness patterns are site-specific, outcome-related, or care-intensity-related. Report the fraction of missingness for every major feature and cohort.

8. Machine Learning, Deep Learning and Statistical Model Program

8.1 Baseline ladder

Every disease-specific experiment should use a pre-specified model ladder so that deep learning is judged against strong conventional baselines rather than assumed to be superior.

Family	Models	Role
Clinical/statistical	Reference ranges, rule-based scores, linear/logistic regression	Essential baseline and interpretability.
Regularized regression	L1/L2/Elastic Net logistic regression	High-dimensional sparse/robust baseline.
Tree ensemble	Random Forest	Nonlinear interactions; robust benchmark.
Gradient boosting	XGBoost, LightGBM, CatBoost	Primary high-performance tabular candidates.
SVM	Linear/RBF SVM	Useful when sample size and feature geometry justify it.
Shallow neural	MLP	Neural tabular baseline.
Temporal neural	GRU/LSTM	Sequential laboratory trajectories.
Transformer	Time-aware Transformer / event transformer	Advanced irregular longitudinal representation.
Survival	Cox / penalized Cox	Interpretable time-to-event baseline.
Survival ML	Random Survival Forest / gradient boosting survival	Nonlinear time-to-event modeling.
Deep survival	DeepSurv or related models	Advanced survival research track.
Anomaly	Isolation Forest, LOF, One-Class SVM	Label-light anomaly detection.
Deep anomaly	Autoencoder, VAE	Latent physiological representation/anomaly score.
Ensemble	Soft voting, weighted voting, stacking	Only after single models are benchmarked and calibrated.
Uncertainty	Conformal prediction, deep ensembles, bootstrap	Decision-support uncertainty and abstention.

8.2 Recommended development sequence

1. Rule-based reference baseline.
2. Logistic regression / regularized regression.
3. XGBoost / LightGBM / CatBoost.
4. Random Forest as a diverse benchmark.
5. Calibrated ensemble.
6. MLP.
7. Temporal GRU/LSTM.

8. Time-aware Transformer.
9. Survival models.
10. Anomaly autoencoder/VAE.
11. Conformal/uncertainty layer.
12. Only then investigate multi-task or foundation-style representations.

8.3 Multi-task learning

A later research track can predict multiple related outcomes from a shared representation, e.g., cardiovascular, renal, metabolic, hepatic, and hematologic endpoints. Multi-task learning should be evaluated against independent task-specific models. Shared representations can be useful but may also cause negative transfer, so task interactions must be measured rather than assumed.

9. Model Evaluation and Statistical Analysis Plan

9.1 Primary performance metrics

- Discrimination: AUROC and area under the precision-recall curve (AUPRC).
- Classification operating points: sensitivity, specificity, PPV, NPV, F1; report confidence intervals.
- Calibration: calibration intercept/slope, calibration plots, and Brier score.
- Clinical utility: decision-curve analysis/net benefit where appropriate.
- Time-to-event: concordance index, time-dependent AUC, calibration at clinically relevant time horizons, and survival curves as appropriate.
- Anomaly detection: precision/recall against reference labels where available; stability across resamples and datasets; false-alert rate per patient or per 1,000 assessments.

9.2 Uncertainty and confidence intervals

Performance should be reported with uncertainty, preferably through bootstrap confidence intervals or another pre-specified approach. Do not rely on a single point estimate. For external validation, report the full confidence interval and calibration behavior, not just AUROC.

9.3 Subgroup evaluation

- Age bands.
- Sex.
- Care setting.
- Dataset/site.
- Baseline disease burden.
- Missingness strata.
- Other ethically and legally appropriate protected or clinically relevant groups available in the source data.

9.4 Model comparison

Use paired patient-level comparisons where suitable, such as bootstrap differences in AUROC/AUPRC, calibration comparisons, and decision-curve comparisons. Do not select a model solely because it has the highest AUROC; calibration, stability, interpretability, computational cost, external validity, and false-alert burden must be considered.

10. Explainability, Uncertainty and Clinical Communication

10.1 Explanation layers

- Data explanation: what data were available and what was missing?
- Trend explanation: which measurements changed, how quickly, and how persistently?
- Model explanation: which features most influenced this particular output?
- Population explanation: how common is this pattern in the training/validation population?
- Uncertainty explanation: how certain is the model and is it inside its validated domain?
- Action explanation: what should a clinician consider reviewing next, and what evidence supports that consideration?

10.2 SHAP and alternatives

SHAP can be used for tree and other compatible models, but explanation stability should itself be evaluated. Do not treat feature attribution as proof of causality. A feature may be predictive because it captures care intensity, disease severity, or other correlated processes. Use permutation importance, counterfactual or sensitivity analyses where appropriate to triangulate explanations.

10.3 Clinical language policy

- Preferred: “elevated risk signal”, “pattern warrants review”, “model identified persistent change”.
- Avoid: “the patient definitely has cancer” or “start treatment now” from laboratory-only AI output.
- Every high-severity alert should include an uncertainty statement and a visible “AI decision support - not a diagnosis” indicator during research use.

11. Federated Learning Research Track

11.1 Why federated learning fits this program

Healthcare data are fragmented across hospitals and health systems. Federated learning (FL) can allow multiple sites to train a shared model without centralizing raw patient records. Recent reviews show rapid growth of healthcare FL, but also emphasize that many studies remain proof-of-concept and that generalization, privacy, communication cost, and methodological quality remain important barriers. [22][23][24]

11.2 Phase-in strategy

1. Synthetic federation: create 3-10 Synthea nodes with controlled differences in age, disease prevalence, missingness, lab units and measurement noise.
2. Benchmark centralized vs federated training under IID data.
3. Introduce non-IID site differences (label prevalence, feature distributions, missingness, lab ordering patterns).
4. Compare FedAvg, FedProx and SCAFFOLD; study personalization when a global model is not optimal.
5. Add secure aggregation and privacy-preserving mechanisms where appropriate.
6. Evaluate communication cost, convergence, fairness across sites, and model drift.
7. Only then seek a real multi-site collaboration governed by data-use agreements and institutional approvals.

11.3 Recommended tooling

Flower is a strong Python-first framework for this track and provides tutorials for building federated applications and customizing strategies. NVIDIA FLARE is another option for more enterprise-oriented federated learning environments. [25]

11.4 Federated evaluation metrics

- Global model AUROC/AUPRC/calibration.
- Per-site performance and confidence intervals.
- Performance dispersion across sites.
- Worst-site and worst-subgroup performance.
- Communication rounds and bytes transferred.
- Model poisoning/robustness tests when appropriate.
- Privacy leakage assessment and threat model documentation.

11.5 Important caution

Federated learning is not automatically private. Gradients or model updates can leak information under some threat models. Therefore the research should define an attacker model and consider secure aggregation, differential privacy, access controls, audit logging, and other safeguards. FL should be treated as a distributed learning architecture, not as a blanket privacy guarantee.

12. Agentic AI Research Track

12.1 Where agents should and should not be used

Agentic AI is appropriate as an orchestration layer around deterministic tools and validated models, for example to retrieve evidence, check data quality, call a risk model, compare prior laboratory reports, generate a draft report, or route a case to a defined workflow. It should not be given unconstrained authority to diagnose, prescribe, change patient records, or bypass model thresholds.

Recent 2026 reviews describe healthcare agent research as promising but immature: a scoping review identified only a small number of rigorously evaluated studies and emphasized the limited clinical validation to date; a broader 2026 survey similarly identifies reliability, privacy, governance, and operational integration as major barriers. [26][27]

12.2 Proposed agent architecture

Agent	Purpose	Control
Data Quality Agent	Checks completeness, units, dates, duplicates, implausible values.	Read-only tools; deterministic rules.
Lab Interpretation Agent	Retrieves standardized definitions and summarizes trends.	LOINC/OMOP metadata + validated calculation library.
Model Orchestrator Agent	Selects which validated model to call based on data coverage and use case.	Allow-list only; no model generation.
Evidence Agent	Retrieves approved clinical sources and documents provenance.	RAG with curated sources and citations.
Safety/Verifier Agent	Checks claims, confidence, contradictions, and forbidden language.	Rule-based + independent verifier.
Report Agent	Produces clinician-readable summary.	Template-constrained generation.
Human Gate	Final approval before a clinical action.	Clinician remains decision-maker.

12.3 Agentic research experiments

1. Can agents correctly route laboratory-only cases to the proper validated model?
2. Can agents retrieve the correct evidence with source citations?
3. Can independent verifier agents reduce unsupported clinical claims?
4. What is the hallucination rate under adversarial and ambiguous laboratory cases?
5. Can a constrained agent reduce documentation time without degrading factual accuracy?
6. How often does the agent abstain appropriately when data are insufficient?

13. End-to-End Technical Architecture

The research platform should separate raw-data ingestion, standardized clinical representation, statistical/ML models, agentic orchestration, and the clinician interface. This separation lets you evaluate the scientific model independently of the interface or LLM layer.

13.1 Proposed architecture

Layer	Function	Suggested implementation
Input layer	PDF/CSV/Excel/FHIR/EHR extracts	Document parser, structured import, optional OCR for scanned documents
Normalization	Units, reference ranges, vocabularies	UCUM/LOINC mapping, OMOP alignment, quality checks
Patient timeline	Longitudinal observations	Temporal indexing, trajectory construction
Feature layer	Current, reference-relative, temporal, derived	Feature registry + versioned transformations
AI layer	Anomaly, classification, survival, temporal DL	scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch
Calibration/uncertainty	Probabilities + abstention	Calibration models, conformal methods, bootstrapping
Explainability	Why this result?	SHAP, feature/trend evidence, model card
Agent layer	Orchestration and reporting	Tool allow-list, RAG, verifier, audit logs
Application	Clinician dashboard and reports	React + TypeScript + FastAPI
Research governance	Provenance, audit, experiment registry	MLflow, DVC, Git, metadata registry

13.2 Python project structure

```

mir-medical-intelligence/
├── README.md
├── pyproject.toml
├── uv.lock / requirements.lock
├── CITATION.cff
├── LICENSE
├── CHANGELOG.md
├── docs/
│   ├── research_protocol.md
│   ├── research_questions.md
│   ├── cohort_definitions/
│   ├── data_sources.md
│   ├── data_dictionary.md
│   ├── model_cards/
│   ├── risk_management.md
│   ├── regulatory_notes.md
│   └── papers/
├── data/          # never commit restricted patient-level data
│   ├── raw/
│   ├── interim/
│   └── processed/

```

- | └─ metadata/
- └─ notebooks/
 - | └─ 01_data_audit/
 - | └─ 02_cbc_baseline/
 - | └─ 03_anomaly_detection/
 - | └─ 04_supervised_benchmarks/
 - | └─ 05_temporal_models/
 - | └─ 06_survival/
 - | └─ 07_external_validation/
 - | └─ 08_federated_learning/
 - | └─ 09_agent_evaluation/
- └─ src/mir_medical/
 - | └─ ingestion/
 - | └─ schemas/
 - | └─ harmonization/
 - | └─ cohort/
 - | └─ features/
 - | └─ anomaly/
 - | └─ models/
 - | └─ temporal/
 - | └─ survival/
 - | └─ calibration/
 - | └─ uncertainty/
 - | └─ explainability/
 - | └─ federated/
 - | └─ agents/
 - | └─ evaluation/
 - | └─ reporting/
- └─ configs/
 - | └─ datasets/
 - | └─ cohorts/
 - | └─ models/
 - | └─ experiments/
- └─ pipelines/
 - | └─ ingest.py
 - | └─ harmonize.py
 - | └─ build_cohort.py
 - | └─ build_features.py
 - | └─ train.py
 - | └─ evaluate.py
 - | └─ generate_report.py
- └─ tests/
- └─ app/
 - | └─ backend/
 - | └─ frontend/
- └─ models/
 - | └─ registry/
 - | └─ checkpoints/
 - | └─ archived/
- └─ reports/
 - | └─ figures/
 - | └─ tables/
 - | └─ experiments/

13.3 Recommended Python libraries

Area	Libraries
Data	pandas, polars, numpy, scipy, pyarrow
Statistics	statsmodels, scipy

Area	Libraries
Classical ML	scikit-learn
Boosting	xgboost, lightgbm, catboost
Deep learning	torch, pytorch-lightning or equivalent
Time series/survival	sktime where appropriate, lifelines, scikit-survival
Explainability	shap
Experiment tracking	mlflow
Data/versioning	dvc or lakeFS, Git
Federated	Flower; NVIDIA FLARE where required
FHIR/clinical data	FHIR client libraries, terminology services; validate against HL7/LOINC/OMOP specifications
API	FastAPI, pydantic
UI	React + TypeScript + Plotly
Testing/quality	pytest, ruff, mypy, pre-commit

13.4 Optional R environment

R should be retained as a secondary statistical environment for biostatistics, survival analysis, epidemiology, sensitivity analyses, and publication-quality statistical graphics. Recommended packages include tidyverse, data.table, survival, tidymodels, rms, caret or modern equivalents, ggplot2, and survey for complex NHANES analyses. The canonical production/research pipeline remains Python to minimize duplication.

14. Core Experimental Program

ID	Experiment	Question	Data	Methods	Output
EXP-001	CBC data audit	What are the distributions, missingness and reference-range patterns?	NHANES	Descriptive stats, QC	Data dictionary + baseline report
EXP-002	Population anomaly detection	Can unsupervised models detect unusual lab profiles?	NHANES	z-score, Mahalanobis, IF, AE	Anomaly benchmark
EXP-003	Clinical abnormality prediction	Can models predict a defined outcome from routine labs?	MIMIC-IV	LR, RF, XGB, LGBM, CatBoost	Benchmark + calibration
EXP-004	Trajectory advantage	Do longitudinal features outperform current values?	MIMIC-IV	Static vs temporal features	Paired model comparison
EXP-005	External validation	Does the locked model transport?	eICU	Frozen model	External validity report
EXP-006	Survival prediction	Can labs estimate 1/3/5-year risk for selected outcome?	All of Us/UKB/N3C as eligible	Cox, RSF, DeepSurv	Time-to-event analysis
EXP-007	Cancer signal	Do labs provide an early risk signal for selected cancer?	All of Us/UKB/N3C as eligible	Temporal + survival ML	Oncology feasibility paper
EXP-008	Federated benchmark	Does FL match centralized training?	Synthetic nodes	FedAvg/FedProx/SCAFFOLD	FL methods paper
EXP-009	Agent routing	Can constrained agents safely orchestrate models?	Synthetic + retrospective cases	Tool-use + verifier	Agent safety report
EXP-010	Silent prospective study	What happens in live workflow without influencing care?	Clinical partner	Prospective silent validation	Deployment readiness report

15. First 90 Days - Concrete Execution Plan

Days 1-15: scientific and data foundation

- Freeze research charter, scope, hypotheses, outcomes, and exclusions.

- Create Git repository and research log template.
- Install Python environment; create project skeleton and test suite.
- Download synthetic Synthea data and build FHIR/CSV import pipeline.
- Download NHANES public CBC and selected laboratory files.
- Write the first data dictionary and unit-normalization registry.

Days 16-30: laboratory representation

- Build canonical laboratory table.
- Implement quality flags and unit normalization.
- Implement reference-range-relative features.
- Create missingness report.
- Create first patient/timepoint and trajectory data structures.
- Document every transformation in code and metadata.

Days 31-60: anomaly research

- Implement robust z-score and Mahalanobis baselines.
- Implement Isolation Forest, LOF and One-Class SVM.
- Implement an autoencoder baseline.
- Compare stability across resampling and population strata.
- Create first anomaly report and visualization layer.

Days 61-90: MIMIC + supervised baseline

- Complete PhysioNet credentialing/CITI/DUA steps and access MIMIC-IV. [1]
- Create an initial cohort for one well-defined clinical outcome.
- Train logistic regression, random forest, XGBoost, LightGBM and CatBoost.
- Set patient-level and temporal validation splits.
- Generate AUROC, AUPRC, calibration, confidence intervals, SHAP and error analysis.
- Write Research Report v0.1 documenting the full experiment.

16. 24-Month Research Roadmap

Period	Focus	Milestone
Months 1-3	Infrastructure + NHANES + Synthea	Canonical schema, data dictionary, anomaly baseline
Months 4-6	MIMIC-IV baseline prediction	First disease-specific benchmark + calibration
Months 7-9	Temporal modeling	Trajectories, GRU/LSTM, survival baseline
Months 10-12	External validation	eICU validation + dataset shift paper
Months 13-15	All of Us / advanced outcomes	Broader population + selected disease/cancer outcomes
Months 16-18	Federated learning	Synthetic non-IID federation + privacy threat model
Months 19-21	Agentic orchestration	Tool-use, RAG, verifier, safety benchmark
Months 22-24	Clinical workflow readiness	Silent prospective design, model cards, risk management, manuscript package

17. Research Paper and Evidence Package Strategy

17.1 Suggested manuscripts

Paper	Topic	Design	Data	Primary contribution
Paper 1	Laboratory-only physiological anomaly detection	Cross-sectional + anomaly methods	NHANES	Descriptive/anomaly validity
Paper 2	Longitudinal laboratory trajectories	Retrospective cohort	MIMIC-IV	Trajectory prediction
Paper 3	External validation and dataset shift	External validation	MIMIC-IV -> eICU	Transportability
Paper 4	Laboratory-only cancer risk signals	Retrospective cohort / survival	All of Us or UKB/N3C if eligible	Oncology feasibility
Paper 5	Federated laboratory risk modeling	Methodological / multi-site simulation	Synthea + partner nodes	Privacy-preserving ML
Paper 6	Constrained agentic clinical intelligence	Benchmark + safety evaluation	Synthetic + retrospective cases	Agent reliability and governance

17.2 Reporting frameworks

Use TRIPOD+AI for prediction model development/evaluation, PROBAST+AI for bias and applicability assessment, and STARD/STARD-AI principles when an experiment evaluates diagnostic accuracy against a reference standard. These frameworks emphasize transparent cohort definition, predictor/outcome handling, preprocessing, model development, validation, fairness, explainability and appropriate interpretation. [11][12][13]

17.3 Research package for every experiment

- Research question and pre-specified hypothesis.
- Data-source/version record.
- Cohort SQL or reproducible cohort code.
- Feature specification and data dictionary version.
- Training/validation/test split definitions.
- Model configuration and random seed.
- Evaluation script and raw metrics output.
- Confidence intervals and calibration plots.
- Subgroup and error analysis.
- Model card and limitations.
- Researcher interpretation and next experiment.

18. Ethics, Privacy, Safety and Regulatory Roadmap

18.1 Research governance

Health data are a special category of personal data under EU data protection law, and scientific research involving such data requires an appropriate lawful basis and safeguards. The EDPB published Guidelines 1/2026 on processing personal data for scientific research, including safeguards for large-scale processing of special-category data and attention to national-law conditions. [18][19]

18.2 EU medical-device implications

The intended use must be documented before software is positioned as a clinical product. Under MDR Rule 11, software intended to provide information used to take diagnostic or therapeutic decisions is generally class IIa, with escalation to class IIb or class III depending on the potential consequences of those decisions. The European Commission updated MDCG 2019-11 on software qualification/classification in June 2025. [16][17]

18.3 EU AI Act status as of 5 September 2026

The EU AI Act generally applies from 2 August 2026, while AI systems embedded in regulated products under Annex I, which can include medical devices, have an extended transition for the high-risk rules to 2 August 2028 under the current text. The legal position should be rechecked at every product milestone because implementation guidance and amendments evolve. [17][18]

18.4 ISO 14971 and lifecycle risk management

ISO 14971:2019 remains current and provides a lifecycle risk-management process covering medical-device software and in-vitro diagnostic software. The research program should create a preliminary hazard log early rather than waiting for commercialization. [17]

18.5 Proposed hazard classes

Hazard	Example	Control
False negative	Model misses a serious risk signal.	Uncertainty, sensitivity-focused thresholds, abstention, clinical review.
False positive	Excessive alerts or unnecessary workup.	Alert-rate monitoring, specificity/PPV, decision curves, workflow limits.
Dataset shift	Model fails at another site.	External validation, site monitoring, recalibration protocol.
Label leakage	Performance inflated by future information.	Index-date firewall, automated temporal checks, clinician review.
Reference-range error	Wrong interpretation due to unit/range mismatch.	Preserve source range, UCUM, validation tests.
Automation bias	Clinician over-trusts AI.	Human factors testing, explanation, uncertainty, explicit AI status.
Agent hallucination	Generated narrative contains unsupported claim.	Evidence citations, verifier agent, constrained templates, human sign-off.
Privacy leakage	Model/data expose sensitive information.	Access controls, encryption, audit logs, DUA, FL/privacy methods where appropriate.

19. Clinician Interface and User Experience

19.1 Core views

- Patient overview: data availability, last laboratory assessment, key active signals.
- Laboratory timeline: trends, abnormality persistence, change points, comparison to prior measurements.
- Risk matrix: disease-system risk signals with calibrated probabilities and confidence/coverage.
- Explanation view: top contributing features, trend evidence, reference ranges, and model version.
- Clinical review: suggested considerations, evidence sources, and “why this was flagged”.
- Audit/research view: dataset, model version, feature version, prediction timestamp, and decision trace.

19.2 Data-coverage indicator

The interface should visibly distinguish well-supported assessments from low-coverage assessments. For example: “Hematology: high coverage; Metabolic: moderate; Oncology: insufficient evidence.” A system should not create the appearance of a full-body medical assessment when only one CBC is available.

19.3 Example report language

Example research-output phrasing: “The model identified a persistent change in hemoglobin and RDW over the available observation window. This pattern is non-specific and does not establish a diagnosis. The result is shown as a risk signal for clinician review. The model was trained on [dataset/version], validated on [dataset/version], and has a calibrated performance of [metric + CI] in the validated cohort.”

20. Optional R Companion Structure

```
mir-medical-r/
├── renv.lock
├── R/
│   ├── cohort_helpers.R
│   ├── survival_models.R
│   ├── survey_weights_nhanes.R
│   ├── calibration.R
│   └── publication_figures.R
├── analysis/
├── reports/
└── tests/testthat/
```

Keep one source of truth for cohort definitions and feature specifications. R should consume exported analysis-ready tables or shared code outputs rather than independently redefining the cohort.

21. Governance, Reproducibility and Model Lifecycle

21.1 Experiment registry

Each experiment receives an immutable ID, dataset version, cohort version, feature version, model version, random seed, code commit hash, configuration file, metrics, and artifacts. MLflow can track runs, parameters and artifacts; Git records code; DVC or equivalent can track data metadata and derived datasets.

21.2 Model cards

Field	Required content
Intended use	What the model is for and what it is not for.
Population	Inclusion/exclusion, demographics, care setting.
Input features	Exact measurements, windows, preprocessing.
Training data	Dataset/version and access restrictions.
Validation	Internal, temporal, external; sizes.
Performance	Discrimination, calibration, uncertainty.
Subgroups	Observed performance differences.
Failure modes	Known false-positive/negative patterns.
Limitations	Generalizability, missingness, confounding, reference standard.
Monitoring	Drift, recalibration, incident response.
Versioning	Model ID, code commit, training timestamp.

21.3 Change control

No model should silently change in production. Any new model or recalibration requires a version increment, validation report, approval record, and documented impact assessment. A model may be retired if calibration, subgroup performance, or data drift exceed pre-defined limits.

22. Research Risks and Mitigation Register

Risk	Impact	Mitigation
Insufficient signal	Low or non-specific laboratory signal for some diseases.	Focus on defined outcomes; publish negative results; use imaging/pathology only as reference standards, not predictors in the lab-first track.
Confounding	Model learns care intensity instead of biology.	Temporal index, causal sensitivity analyses, site validation, clinical review.
Spectrum bias	Training population differs from target population.	Multiple datasets and external validation.
Class imbalance	Rare outcomes distort metrics.	AUPRC, class-weighting, focal loss where justified, sampling only inside training.
Overfitting	Too many features and experiments.	Pre-specification, nested CV where needed, untouched test set, model simplicity baselines.
Dataset shift	Performance collapses across sites.	External validation and federated/site-aware methods.
Agent overreach	LLM invents unsupported medical claims.	Tool allow-list, evidence citations, verifier, hard refusal/abstention.
Regulatory mismatch	Research prototype accidentally marketed as clinical device.	Clear research labeling, intended-use control, regulatory review before deployment.
Privacy breach	Improper use or release of restricted data.	DUAs, secure storage, access control, no raw-data commits, privacy review.

23. Success Criteria

23.1 Scientific success

- At least one laboratory-only anomaly model demonstrates reproducible detection against a defensible reference definition.
- At least one longitudinal model shows whether historical trajectories add predictive value over cross-sectional features.
- At least one model is externally validated with calibrated performance and documented transportability limitations.
- At least one cancer-risk question is tested using a pre-specified outcome window and leakage-controlled cohort.
- Federated experiments quantify the trade-off between centralized and decentralized training.
- Agent experiments quantify factuality, abstention, routing accuracy, and safety rather than subjective “looks good” judgments.

23.2 Engineering success

- Every model result is reproducible from a versioned configuration.
- Dataset ingestion and feature engineering have automated tests.
- The clinical interface displays model provenance and uncertainty.
- The research platform can run on synthetic data without restricted access.
- No restricted dataset is accidentally committed to source control or shared outside its permitted environment.

24. Immediate Acquisition Checklist

Step	Action	Deliverable
1	Create research identity/account	PhysioNet, All of Us, relevant institutional accounts.
2	Complete CITI training	Required for MIMIC/eICU; retain certificates and dates.
3	Execute MIMIC/eICU DUA	Follow current PhysioNet process.
4	Download NHANES files	Start with CBC_L and related laboratory files; preserve documentation.
5	Install synthetic environment	Run Synthea locally and generate FHIR/CSV test cohorts.
6	Assess All of Us eligibility	Check whether your institution has a current DURA and which tier is available.
7	Prepare UKB concept note	Develop a concise scientific proposal for the late-2026 application window if needed.
8	Define first outcome	Choose one common, well-labeled clinical endpoint for MIMIC-IV.
9	Create protocol before modeling	Freeze index date, outcome, predictors, split strategy and primary metrics.
10	Start experiment EXP-001	Population laboratory audit + data dictionary + anomaly baseline.

25. Reference Sources and Access Links

- [1] PhysioNet. MIMIC-IV v3.1; credentialed access, CITI training, DUA. [Source](#)
- [2] PhysioNet / MIT LCP. eICU Collaborative Research Database v2.0; credentialed access, training and DUA. [Source](#)
- [3] CDC/NCHS. NHANES Continuous Laboratory Data 2021-2023. [Source](#)
- [4] NIH All of Us. Data access tiers and Researcher Workbench. [Source](#)
- [5] NIH All of Us. Researcher Workbench and current data collections. [Source](#)
- [6] NIH All of Us. Data sources and OMOP standardization. [Source](#)
- [7] CDC/NCHS. CBC_L documentation, 2021-2023. [Source](#)
- [8] UK Biobank. Apply for access; current 2026 application status. [Source](#)
- [9] UK Biobank Community. How to complete an access application. [Source](#)
- [10] NCATS NIH. N3C data overview and access requirements. [Source](#)
- [11] BMJ. TRIPOD+AI reporting guideline for prediction models using regression or ML. [Source](#)
- [12] BMJ. PROBAST+AI: quality, risk of bias and applicability. [Source](#)
- [13] BMJ Open. STARD-AI protocol and diagnostic-accuracy reporting considerations. [Source](#)
- [14] Synthea. Synthetic Patient Population Simulator and SyntheticMass. [Source](#)
- [15] NCI SEER. SEER incidence data and cancer registry resources. [Source](#)
- [16] European Commission / MDCG. MDCG 2019-11 rev.1: software qualification and classification, June 2025. [Source](#)
- [17] EUR-Lex. Regulation (EU) 2017/745, including MDR Rule 11. [Source](#)

- [18] European Commission. EU AI Act official regulatory framework and timeline. [Source](#)
- [19] EDPB. Guidelines 1/2026 on processing personal data for scientific research. [Source](#)
- [20] LOINC. About LOINC and laboratory/clinical observation identifiers. [Source](#)
- [21] OHDSI. OMOP Common Data Model. [Source](#)
- [22] PubMed. Federated machine learning in healthcare: systematic review. [Source](#)
- [23] PubMed. Implementing federated learning in healthcare: pitfalls and recommendations. [Source](#)
- [24] PubMed. Methodological gaps limiting federated learning in healthcare, 2026. [Source](#)
- [25] Flower. Federated learning framework and tutorials. [Source](#)
- [26] PubMed. The role of agentic artificial intelligence in healthcare: scoping review, 2026. [Source](#)
- [27] PubMed. A comprehensive survey of AI agents in healthcare, 2026. [Source](#)
- [28] WHO. Regulatory considerations on artificial intelligence for health. [Source](#)
- [29] WHO. Ethics and governance of AI for health / large multi-modal models. [Source](#)
- [30] ISO. ISO 14971:2019 medical-device risk management; confirmed current in 2025. [Source](#)

Appendix A. Example Research Log Template

Experiment ID:
 Date:
 Research question:
 Hypothesis:
 Dataset + version:
 Access/DUA reference:
 Cohort definition:
 Index date definition:
 Outcome definition:
 Predictor window:
 Excluded information / leakage controls:
 Preprocessing version:
 Feature version:
 Models:
 Hyperparameters/config:
 Split strategy:
 Primary metric:
 Secondary metrics:
 Calibration method:
 Confidence interval method:
 Subgroups:
 Results:
 Error analysis:
 Limitations:
 Interpretation:
 Decision: keep / revise / reject:
 Next experiment:
 Code commit:
 Model version:

Appendix B. Minimum Model Card Template

Model name/version: _____

Intended use: _____

Not intended for: _____

Training population: _____

Features: _____

Training dataset/version: _____

Internal validation: _____

External validation: _____

AUROC/AUPRC: _____

Calibration: _____

Known failure modes: _____

Fairness/subgroups: _____

Uncertainty/abstention: _____

Monitoring plan: _____

Clinical/regulatory status: research only / under evaluation / approved use (must be documented)

Appendix C. Research Position Statement

The central scientific opportunity is not to prove that a CBC can diagnose every disease. It is to measure, rigorously and honestly, how much information routine non-imaging laboratory data contain about physiological state and future disease risk, how much of that information persists across time and across institutions, and whether models can convert that signal into calibrated, explainable, clinician-useful decision support. A successful program may discover strong signals for some outcomes, weak signals for others, and no meaningful signal for still others. The methodology is designed to make all three outcomes useful scientific findings.